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ORIGINAL INVESTIGATION

Reliability of Three Landmine-Punch-Throw Variations and Their Load–Velocity Relationships Performed With the Dominant and Nondominant Hands

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Purpose: This study assessed the reliability and load–velocity profiles of 3 different landmine-punch-throw variations (seated without trunk rotation, seated with trunk rotation, and standing whole body) with different loads (20, 22.5, and 25.0 kg), all with the dominant hand and nondominant hand. ***Methods:*** In a quasi-randomized order, 14 boxers (24.1 [4.3] y, 72.6 [10.1] kg) performed 3 repetitions of each variation with their dominant hand and their nondominant hand, with maximal effort and 3 minutes of interset rest. Peak velocity was measured via the GymAware Power Tool (Kinetic Performance Technologies). The interclass correlation coefficients and their 95% CIs were used to determine the intrasession reliability of each variation × load × hand combination. Additionally, a 2 (hand) × 3 (variation) repeated-measures analysis of variance assessed the load–velocity profile slope, and a 3 (variation) × 2 (hand) × 3 (load) repeated-measures analysis of variance assessed the peak velocity of each variation. ***Results:*** Most variations were highly reliable (intraclass correlation coefficient > .91), with the nondominant hand being as reliable or more reliable than the dominant hand. Very strong linear relationships were observed for the group average for each variation ($R^2 \geq .96$). However, there was no variation × hand interaction for the slope, and there was no main effect for variation or hand. Additionally, there was no interaction for the peak velocity, but there were main effects for variation, hand, and load ($P < .01$). ***Conclusion:*** Each variation was reliable and can be used to create upper-body ballistic unilateral load–velocity profiles. However, as with other research on load–velocity profile, individual data allowed for more accurate profiling than group average data.

Keywords: resistance training, combat sports, boxing, ballistic medicine-ball throw, bench throw

There are numerous methods of assessing muscular strength, such as repetition maximum tests,[1] isokinetic strength,[2–4] and others.[5] Although a large selection of strength tests exists, testing ballistic power output is largely limited to movements that encompass jumps[6,7] and throws.[8–10] Indeed, jump testing is widely used, especially for most athletes who perform jumps during training, which likely reduces the variability and need for further familiarization prior to testing. However, jumps are essentially limited to the lower body, necessitating similar solutions for upper body power assessments. To test upper body power output, 1 common choice includes throwing a medicine ball for distance,[11] but the resultant data can largely depend on the throwing technique and size of the implement, variations of which may result in large variability and unreliable test results. It is true that bench press throws include fewer degrees of freedom, especially if performed on a Smith machine, for example, which should result in more reliable data. However, bilateral exercises cannot always be used for testing unilateral movements,[12] which might be desired for specific purposes such as quantifying asymmetries,[13–15] assessing training adaptations between limbs,[16] or performing exercises where each limb may necessitate a different loading pattern.

The landmine punch throw is a fairly novel unilateral ballistic upper body exercise that is commonly used not only in training but also for testing. During the landmine punch throw, an athlete grabs the end of 1 barbell sleeve and throws it with a linear upward push (approximately 40°–60° from parallel), while the other sleeve (ie, the opposite end of the barbell) is fixed to a 3-dimensional moveable attachment on the floor.[17] As such, this exercise allows for upper body unilateral ballistic testing, requires minimal equipment, and is extremely portable. The movement is often performed in a standing position using the whole body, but different variations of the landmine punch throw can allow for isolated testing of the upper body, upper body and trunk, and the whole body, including the lower limbs. Along these lines, it is possible that the different involvement of body segments will alter the characteristics of the load–velocity profile of the landmine punch throw, as is, for example, affecting peak velocity (PV) in a real boxing punch.[18] In terms of testing, a linear position transducer can be attached to the thrown end of the barbell to assess PV, and the sleeve can be loaded to assess a wider range of external forces, which could ultimately lead to the creation of individualized upper body ballistic load–velocity profiling. In fact, using a unilateral load–velocity profile could be very useful for exploring asymmetry between limbs. Additionally, this load–velocity profile could also be useful to track adaptations over time, specifically when using different body segments.

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In practice, the landmine punch throw is already used in training and testing for sports that share similar movement patterns, such as combat sports, rugby, American football, and other sports where the arm and hand require rapid extension in front of the body. Next to that, landmine push throw without throwing has been investigated, especially its load–velocity profile and reliability.[19,20] However, to the best of our knowledge, the reliability of the landmine punch throw test in addition to the load–velocity profile derived from the landmine punch throw have not been scientifically addressed in the literature. Specifically, the reliability of different variations of the exercise performed with different loads with the dominant hand (DH) and nondominant hand (NH), and their load–velocity relationships, are all some of the foundational points that should be addressed before promoting the widespread use of testing procedures that may be unreliable.

Therefore, 1 aim of the present study was to determine the PV reliability of the 3 independent variations of the landmine punch throw (whole body landmine punch throw [LP], landmine punch throw in a seated position with trunk rotation [LPw], and landmine punch throw in seated position without trunk rotation [LPwo]), each with 3 different loads (only barbell [20], 22.5, and 25 kg) with the DH and NH. Additionally, another purpose was to evaluate the differences in the characteristics of load–velocity profiles of 3 landmine punch throw variations using 3 different loads with the DH and NH. According to previous studies on a variety of exercises,[20–23] it was hypothesized that landmine punch throw PV would result in at least moderate reliability. Additionally, according to a previous study,[24] it was hypothesized that the velocity intercept would significantly differ between variations and hands. The same was expected for the load intercept and the area under the line of the load–velocity profiles.

Methods

Experimental Approach to the Problem

During a single laboratory visit, 14 trained boxers performed, in a quasi-randomized order, the 3 different landmine punch throw variations with 3 different loads, all with the DH and the NH. The PV of the landmine punch throws was assessed using a linear position transducer.

Sample Size

To calculate the sample size required to estimate the intraclass correlation coefficients (ICCs) of .85 with the lower bound of 95% CI > 0.50 (the threshold for moderate reliability), we used Zou's formula as implemented in *ICC.Sample.Size* package (version 1.0.) in R.[25] Assuming 3 ratings per subject, a desired power of 80%, and using a 2-sided .05 significance level, the required sample size is 12. To account for possible drop-outs or technical issues, we required 14 subjects.

Subjects

All 14 healthy boxers (24.1 [4.3] y, 72.5 [10.1] kg, 176.9 [8.3] cm, 12 orthodox, and 2 southpaw boxers) had at least 1 competitive boxing bout and at least 1 year of structured strength and conditioning training during which they regularly performed the landmine punch throw exercise. Each subject was informed of the potential risks and possible benefits of this project and then read and signed a written informed consent approved by the local university ethics committee (ER19357858).

Procedures

All procedures were performed during 1 testing visit and consisted of 3 phases: (1) warm-up, (2) individual setup and familiarization, and (3) landmine punch throw assessment.

Warm-Up

The standardized warm-up included 120 seconds of rope skipping, mobilization exercises for the upper limbs, lower limbs, and hips, and dynamic stretching for the upper and lower body for 10 minutes, which was followed by 6 squat jumps and 6 countermovement jumps.

Individual Setup and Familiarization

The landmine punch throw was performed in 3 different conditions: LP, LPw, and LPwo, all of which were performed with the both DH and NH independently. In the standing position, each subject stood in their preferred boxing stance (orthodox or southpaw). In the seated position, the seated height of the subject was adjusted with jerk blocks to ensure a 90° knee joint angle. A hand-operated Goniometer was used to determine knee joint angle for each subject. Then, subjects fully extended their legs and rested their heels on a slightly elevated surface to minimize the use of the lower limbs during the movement. The proper technique was demonstrated for each variation. Before each testing set, subjects performed 3 trials of landmine punch throw for each variation and load, with an estimated 50% maximal effort, using the same load as the subsequent test.

Landmine-Punch-Throw Assessment

Each subject performed 3 repetitions of the landmine punch throw variations for each hand with 3 loads (20, 22.5, and 25 kg), with 3 minutes of interset rest. The testing loads were set up based on pilot testing that showed that greater differences between each load did not allow participants to perform the seated variations correctly. Furthermore, the lightest load of 20 kg was chosen because the large majority of landmine base equipment is built to fit the sleeve of a standard 20 kg barbell, making lighter loads not ecologically valid and not possible for the majority of users. A constant time of 3 seconds was provided between each repetition. The initial position for the LP was similar to a true boxing stance. The barbell was held in the rear hand as close as possible to shoulder height, with the elbow fully flexed and the knees slightly flexed. The lead hand was positioned at the chin with the elbow flexed. Upon instruction, subjects proceeded to rotate their trunk on the rear side from a stationary position into a squat before forcefully extending the ankle, knee, hip, and elbow while simultaneously throwing the barbell in a forward direction.[17] The LPw was performed with the same initial position of the lead and the rear hand and the same technique (ie, with rotation of the upper body but now without lower body involvement). The LPwo was the same as the LPw, but a broomstick was positioned behind the back of each participant to avoid the rotation of the trunk (Figure 1). The subjects were required to maintain the same level of contact with the broomstick throughout the movement in order to minimize the occurrence of trunk rotation. Each variation was performed with the DH and NH.

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Figure 1 — Landmine-punch-throw variations: (A) landmine punch throw, (B) landmine punch throw with rotation of the trunk, and (C) landmine punch throw without rotation.

Data Acquisition and Data Analyses

Data Acquisition

The PV of all variations of landmine punch throw and different loads was collected with a validated linear position transducer (GymAware Power Tool, Kinetic Performance Technologies).[26] The cable of the GymAware was attached to the barbell on the internal side, as close as possible to the collar, on the side where the participants held the barbell. The obtained data from the GymAware were transmitted via Bluetooth to a tablet (iPad, Apple, Inc.) using the GymAware v2.4.1 app and to the online cloud before being exported to Microsoft Excel (Microsoft Corporation) for future analysis.

Data Analysis

The intrasession reliability of each variation–load–hand combination was determined by ICCs (A, 1),[27] with their 95% CIs using the irr package (version 0.84.1) in R, version 3.5.2. The coefficient of variation was calculated using the mean squared error method as implemented in the SimplyAgree package (version 0.1.2).[28] The standard error of measurement was calculated using the consecutive pairwise difference method with CIs based on the chi-squared distribution as implemented in the rel package (version 1.4.2).[29] The magnitude of the ICC was interpreted based on its lower bound of the 95% CI (LCI) as follows: <0.50, poor reliability; 0.50 to 0.75, moderate reliability; 0.75 to 0.90, good reliability; and >0.90, excellent reliability.[30] The relationship between PV and the prescribed loads was established via a linear regression using Microsoft Excel (Microsoft Corporation).

The data are presented as means and standard deviations. Individual 2 (hand [DH and NH]) × 3 (variation [LPwo, LPw, and LP]) repeated measures analysis of variance (ANOVA) with a Bonferroni post hoc test was applied to compare the slope and velocity intercept, load intercept, and area under the line of the regression lines of each variation and hand. Last, individual 2 (hand [DH and NH]) × 3 (variation [LPwo, LPw, and LP]) × 3 (load [20, 22.5, and 25 kg]) repeated measures ANOVA with a Bonferroni post hoc test was also applied to compare the PV attained at each variation, hand, and load. Statistical significance was set at an alpha level of $P < .05$, whereas this part of statistical analysis was performed using the software package JASP (Jeffrey's Amazing Statistics Program, version 0.16.2).[31] Hedges g effect sizes with 95% CIs were used to determine the magnitude of the difference between LPwo, LPw, and LP for the slope, velocity intercept, load intercept, and area under the line of linear regression, and were interpreted for highly trained subjects as follows: trivial, $g < 0.25$; small, $g = 0.25$ to 0.50; moderate, $g = 0.50$ to 1.00; and large, $g > 1.00$.[32] Hedges g was computed as an unbiased version of Cohen d, adjusting for the small sample size.[33]

Results

The PVs attained against each load, variation, and hand, with their effect size, are shown in Table 1. The reliability results of each variation–load–hand combination are shown in Table 2. Based on the LCI, most variations displayed good reliability (LCI ranking from 0.75 to 0.89 for individual variations), especially for the NH (LCI = 0.76–0.89), with a few demonstrating moderate reliability

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Table 1 Peak Velocities (m/s; mean [SD]) Attained Against Each Load Between the Landmine-Punch-Throw Variations Performed with the Dominant and Nondominant Hands

Condition	Load (kg)	Dominant hand[a]	Nondominant hand	Hedges *g* (95% CI)
LP[b]	20[c]	2.66 (0.32)	2.54 (0.36)	0.34 (–0.40 to 1.09)
	22.5[d]	2.53 (0.31)	2.43 (0.34)	0.30 (–0.45 to 1.04)
	25	2.36 (0.34)	2.28 (0.32)	0.24 (–0.51 to 0.98)
LPw[e]	20[c]	2.03 (0.14)	1.94 (0.18)	0.54 (–0.21 to 1.30)
	22.5[d]	1.85 (0.17)	1.74 (0.20)	0.58 (–0.18 to 1.33)
	25	1.65 (0.24)	1.58 (0.22)	0.30 (–0.45 to 1.04)
LPwo	20[c]	1.75 (0.19)	1.68 (0.23)	0.32 (–0.42 to 1.07)
	22.5[d]	1.53 (0.20)	1.49 (0.23)	0.18 (–0.56 to 0.92)
	25	1.42 (0.18)	1.27 (0.24)	0.69 (–0.08 to 1.45)

Abbreviations: LP, landmine punch throw; LPw, LP with rotation of the trunk; LPwo, LP without rotation. Note: Hedges *g* was computed as an unbiased version of Cohen *d*, adjusting for the small sample size.
[a]Significantly greater than nondominant hand ($P_{bonf} < .001$). [b]Significantly greater than LPw and LPwo ($P_{bonf} < .001$). [c]Significantly greater than 22.5 and 25 ($P_{bonf} < .001$). [d]Significantly greater than 25 ($P_{bonf} < .001$). [e]Significantly greater than LPwo ($P_{bonf} < .001$).

Table 2 Reliability of Each Variation–Load–Hand Combination of the LP

	Dominant hand							Nondominant hand						
Condition (kg)	ICC	LCI	UCI	SEM	LCI	UCI	CV	ICC	LCI	UCI	SEM	LCI	UCI	CV
LP 20	.89	.75	.96	.10	.07	.14	.04	.93	.84	.97	.09	.07	.13	.04
LP 22.5	.94	.85	.98	.07	.05	.09	.03	.94	.84	.98	.08	.06	.11	.03
LP 25	.77	.54	.91	.16	.12	.22	.06	.89	.76	.96	.10	.08	.15	.05
LPw 20	.88	.75	.96	.05	.04	.08	.03	.89	.77	.96	.07	.05	.10	.03
LPw 22.5	.82	.63	.93	.07	.05	.10	.04	.95	.89	.98	.04	.03	.06	.03
LPw 25	.93	.84	.97	.05	.04	.07	.04	.92	.81	.97	.06	.05	.09	.04
LPwo 20	.87	.72	.95	.06	.04	.08	.04	.94	.87	.98	.05	.04	.07	.04
LPwo 22.5	.95	.89	.98	.04	.03	.06	.03	.92	.83	.97	.06	.05	.09	.05
LPwo 25	.78	.55	.92	.08	.07	.12	.07	.97	.94	.99	.04	.03	.06	.03

Abbreviations: CV, coefficient of variation; ICC, intraclass correlation coefficient; LCI, lower CI; LP, landmine punch throw; LPw, LP with rotation of the trunk; LPwo, LP without rotation; SEM, standard error of measurement; UCI, upper CI.

(LCI = 0.54–0.72) for the DH. In general, the variations performed with the NH were as reliable, or more reliable, than with the DH.

Very strong linear relationships were observed for group averages for LPwo, LPw, and LP performed by DH ($R^2 = .96$, .99, and .99), and NH for each variation ($R^2 = .99$).

The slopes of the linear regression with their effect sizes are shown in Table 3. The 2-way repeated measures ANOVA indicated that there was no variation × hand interaction for the slope of regression lines ($P = .221$, $\eta_p^2 = .111$), and there was no main effect for variation ($P = .098$, $\eta_p^2 = .168$) or hand ($P = .972$, $\eta_p^2 = .0001$).

The velocity and load intercepts of the linear regression with their effect sizes are shown in Table 3. The 2-way repeated measures ANOVA indicated that there was no variation × hand interaction for the velocity intercept of regression lines ($P = .233$, $\eta_p^2 = .107$), and there was no main effect for variation ($P = .103$, $\eta_p^2 = .167$) or hand ($P = .490$, $\eta_p^2 = .037$). Additionally, there was no variation × hand interaction for the load intercept of regression lines ($P = .624$, $\eta_p^2 = .021$), and there was no main effect for variation ($P = .101$, $\eta_p^2 = .189$) or hand ($P = .870$, $\eta_p^2 = .002$).

The area under the line with effect sizes are shown in Table 3. There was no variation × hand interaction for the area under the line ($P = .596$, $\eta_p^2 = .024$), and there was no main effect for hand ($P = .628$, $\eta_p^2 = .019$). However, there was a main effect for variation ($P = .005$, $\eta_p^2 = .426$). Post hoc testing showed that the area under the line of LP was greater than LPw (mean difference = 24.343 [2.836–45.849], $P_{bonf} = .023$, $g = 0.61$ [0.08–1.15]) and LPwo (mean difference = 36.195 [14.688–57.701], $P_{bonf} < .001$, $g = 0.93$ [0.38–1.48]). However, the area under the line of LPw was not significantly different than LPwo (mean difference = 11.852 [–9.655 to 33.358], $P_{bonf} = .511$, $g = 0.83$ [0.28–1.37]).

Additionally, there were no variation × hand × load, variation × hand, variation × load, or hand × load interactions for PV ($P = .128$, $\eta_p^2 = .126$; $P = .932$, $\eta_p^2 = .005$; $P = .093$, $\eta_p^2 = .140$; $P = .737$, $\eta_p^2 = .023$, respectively). However, there was a main effect for variation, load, and hand ($P < .001$, $\eta_p^2 = .934$; $P < .001$, $\eta_p^2 = .941$; and $P = .006$, $\eta_p^2 = .451$), respectively. Post hoc testing showed that the PV of LP was greater than LPw (mean difference = 0.666 [0.523–0.809], $P_{bonf} < .001$, $g = 2.21$ [1.27–3.15]) and LPwo (mean difference = 0.940 [0.767–1.113], $P_{bonf} < .001$, $g = 3.02$ [1.93–4.10]), and PV of LPw was less than LPwo (mean difference = 0.274 [0.191–0.357], $P_{bonf} < .001$, $d = 1.05$ [0.26–1.84]). Additionally, PV was greater with 20 kg than 22.5 kg (mean difference = 0.171 [0.132–0.211], $P_{bonf} < .001$, $g = 0.36$ [–0.38 to 1.11]) and 25 kg (mean difference = 0.338 [0.282–0.394], $P_{bonf} < .001$, $g = 0.71$ [–0.05 to 1.47]), and PV of 22.5 kg was greater than 25 kg (mean difference = 0.167 [0.126–0.207], $P_{bonf} < .001$,

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Table 3 Results of the Slope of Linear Regression With Their SDs

Condition	Dominant hand	Nondominant hand	Hedges *g* (95% CI)
Slope of linear regression			
LP	–0.06 (0.05)	–0.05 (0.04)	0.21 (–0.53 to 0.96)
LPw	–0.08 (0.04)	–0.07 (0.02)	0.31 (–0.44 to 1.05)
LPwo	–0.07 (0.02)	–0.08 (0.02)	0.49 (–0.27 to 1.24)
Velocity intercept of linear regression			
LP	3.88 (1.00)	3.56 (0.93)	0.32 (–0.42 to 1.07)
LPw	3.54 (0.69)	3.35 (0.37)	0.33 (–0.41 to 1.08)
LPwo	3.07 (0.40)	3.32 (0.46)	0.56 (–0.19 to 1.32)
Load intercept of linear regression			
LP	58.92 (44.66)	53.90 (28.03)	0.13 (–0.61 to 0.87)
LPw	42.87 (12.21)	47.99 (7.00)	0.50 (–0.25 to 1.25)
LPwo	42.97 (7.03)	40.70 (5.23)	0.36 (–0.39 to 1.10)
Area under the line of linear regression			
LP	108.81 (60.72)	96.65 (42.68)	0.22 (–0.52 to 0.97)
LPw	77.14 (20.88)	79.63 (8.65)	0.15 (–0.59 to 0.89)
LPwo	66.07 (13.70)	66.99 (10.0)	0.07 (–0.67 to 0.82)

Abbreviations: LP, landmine punch throw; LPw, LP with rotation of the trunk; LPwo, LP without rotation. Note: Hedges *g* was computed as an unbiased version of Cohen *d*, adjusting for the small sample size.

$g = 0.34$ [–0.40 to 1.09]). Last, the PV was greater with DH than NH (mean difference = 0.091 [0.031–0.151], $P_{bonf} = .006$, $g = 0.20$ [–0.18 to 1.33]).

Discussion

This study was designed to determine the PV reliability of the 3 variations of the landmine punch throw performed with different loads and DH and NH. Additionally, this study evaluated the load–velocity profile of each variation and DH and NH. In the current study, the PV achieved was affected by variation, hand, and load. Regardless of the specific variation, load, or hand used, most combinations of the landmine punch throw achieved good reliability (LCI = 0.75–0.89). When considering both the DH and NH, the goodness of fit demonstrated similarity for the group average of each variation of landmine punch throw. Additionally, neither the slope nor the velocity or load intercept of the regression line have not been affected by the hand or variation of the landmine punch throw. However, the area under the line was affected by the variation. Although some studies have determined the load–velocity reliability of upper body bilateral pushing exercise,[34–36] to the best of the authors' knowledge, this is the first study to determine the reliability of upper body unilateral ballistic exercises that can be used for field testing and monitoring.

As expected, the more body segments that were involved in the landmine punch throw, the greater the resultant PV was in the present study. In a similar fashion, others have found the same pattern during punching with the whole body, with the legs fixed, and with the legs and trunk fixed.[18] Another expected outcome was that PV decreased as the load increased, which abides by the inverse load–velocity relationship.[37–39] Additionally, PV was greater with the DH compared to the NH, which is similar to previous research assessing punch velocity.[40]

This aligns with a previous study in which the landmine push throw demonstrated moderate reliability for the right hand (ICC = .73) and excellent reliability for the left hand (ICC = .92) at 20 kg. Moreover, it showed excellent reliability for the right hand (ICC = .90) and good reliability for the left hand (ICC = .81) at 25 kg.[20] The first hypothesis was confirmed. In this study, the most possible combinations of the landmine punch throw were reliable (LCI = 0.75–0.89), regardless of the variation, load, or hand, meaning that all of the different landmine punch throws performed could be used in practice. However, it was interesting that trials performed with the NH were more reliable than those performed with the DH. This may have occurred because PV was greater with the DH, indicating that perhaps the fastest DH trial may have been performed with greater PV than the other repetitions, with the NH moving at a slower speed but more consistently. In support of this idea, previous research showed similar results for peak force in trained boxers, where the NH was more reliable (ICC = .89) than the DH (ICC = .73).[41] In a real-life boxing bout, boxers most often perform straight punches with their lead hand, which is often the NH,[42,43] which might influence the lower variability of the nondominant and reflect greater within-session reliability of NH compared to DN straight punches.[41] Considering that the landmine punch throw used in this study was performed in different variations with the DH and NH and with different loads, it seems that each of those variations could be used as a reliable unilateral ballistic test as part of an upper body force velocity profile.[44–46]

Each landmine punch throw variation performed with the DH and NH had a very strong linear relationship with the slopes of the regression lines, which were not affected by hand or variation. In a previous study by Balsalobre-Fernández et al,[24] the slopes of the regression lines in bilateral and unilateral knee extensions were similar to those of the present study, but that study did not indicate whether differences existed between bilateral and unilateral knee extensions or between the dominant leg and nondominant leg. In resistance training exercises, the velocity typically demonstrates a linear decrease as the load increases.[46] This inverse relationship between velocity and load is an important factor in understanding an individual's performance as it can shed light on the neuromuscular capacities of the athlete accompanied with its alternations

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after a training period.[47] However, the velocity intercept, load intercept, and area under the line, which represent the baseline velocity when the load is 0, the baseline load when the velocity is 0, and power capacities, respectively, can also provide valuable information about a person's performance.[48] Analyzing both the velocity–load relationship and its characteristics can help develop a comprehensive understanding of an individual's performance capabilities in resistance training exercises. In our study, the velocity and load intercepts of the regression lines were not affected by hand or variation, but in the aforementioned study, the velocity intercept of the regression line was significantly different between the bilateral and unilateral knee extensions. However, the area under the line was affected by variation, especially by LP, compared to LPw and LPwo. The effect was influenced by the achieved resultant PV during landmine punch throw variation. The area under the line is calculated as the product of the load intercept multiplied by the velocity intercept divided by 2.[48] In our study, the resultant PV was influenced by the body segments involved, using the LP as an example, achieved greater resultant PV compared to LPw or LPwo.

Additionally, the load–velocity relationship has been explored for a wide range of exercises, such as the bench press throw,[8] deadlift,[1,38] back squat,[49] and others.[21,50] However, no previous research has compared the linearity of the load–velocity relationship for an upper body unilateral ballistic exercise. Our results provided a fairly linear velocity relationship ($R^2 \geq .96$), similar to the bench press throw ($R^2 = .979$),[8] which is commonly used as an upper body bilateral ballistic test. However, bilateral testing cannot observe the asymmetry of the upper body, which makes the landmine punch throw an interesting option for athletes that perform sport actions 1 limb at a time. For example, 2 individual sets of data from the current study (Figure 2) show distinct differences between limbs within 1 fairly untrained subject, while a well-trained subject displayed little-to-no between-limb asymmetry. Therefore, considering the findings of this paper, in addition to the data in Figure 2, using a unilateral test could help identify asymmetries that could not be identified using a bilateral test. Furthermore, putting asymmetries aside, the landmine punch throw can also be used to track an individual's progress over time thanks its reliability, linearity, and goodness of fit. Additionally, further insight into the neuromuscular capabilities of the athlete can be assessed with velocity and load intercepts as well as the area under the curve.[48] These parameters could be used for optimizing training, monitoring performance progress in real-life sports scenarios, identifying asymmetries, and implementing targeted interventions for performance enhancement. However, it is important to consider measurement error and variability that could occur within the landmine punch throw. The standard error of measurement and coefficients of variation are shown in Table 2. Thus, using this upper body unilateral ballistic exercise could be suitable for different sports discipline, where the movement is commonly performed unilaterally, such as punching or throwing.

Practical Application

The landmine punch throw can be used as a reliable upper body unilateral ballistic test for athletes. By performing the test with the DH and NH with different loads (20, 22.5, and 25 kg), the different landmine punch throw variations assessed in this study can all be used to create upper body unilateral load–velocity profiles. However, as with other exercises like the back squat, bench press, etc., the group's average results should not be used as a benchmark for each athlete, which requires load–velocity profiles to be compared within each athlete individually.

Conclusion

All of the landmine-punch-throw loads and variations were reliable for both hands. Therefore, from a practical point of view, the landmine punch throw could be used as a reliable movement-specific test for upper-body ballistic strength and power. However, although the nondominant hand (NH) seemed to result in more reliable data than the dominant hand (DH), future research should confirm whether this holds true for boxers of different calibers, training experiences, weight categories, and the like. Additionally, the slope of the group average regression lines was similar for the

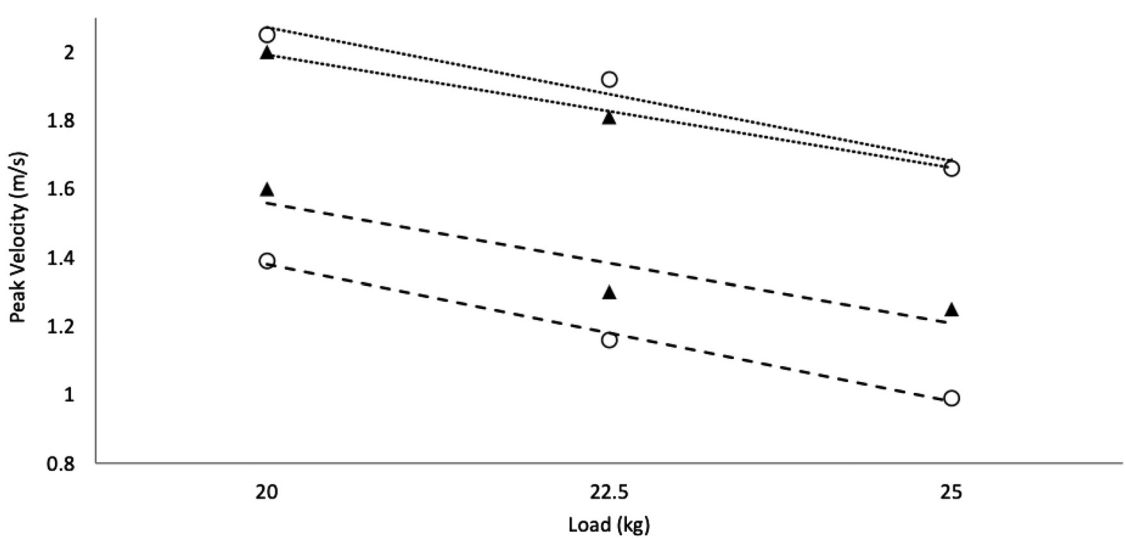

Figure 2 — Example of the load–velocity profile relationship for a landmine punch throw without rotation for a fairly untrained (dashed line) and well-trained subject (dotted line). Circle indicates nondominant hand; triangle, dominant hand.

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LPwo, LPw, and LP for both the DH and NH. Although the group averages showing a very strong linear relationship, the individual data are not as clear (similar to traditional resistance-training exercises like the bench press and back squat) (Figure 2). However, it is important to consider that the difference between loads was 2.5 kg, meaning that it is possible that using a different range of loads during each landmine-punch-throw variation may influence the load–velocity profile, but future research is needed to investigate this hypothesis.

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